08 Support Vector Machines

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Use of SVM

Small to medium-sized non-linear datasets ($[10^2; 10^5]$ instances), high-dimensional linear datasets.

- Linear classification
- Non-linear classification
- Regression (SVR)
- Novelty detection

Linearly separable data

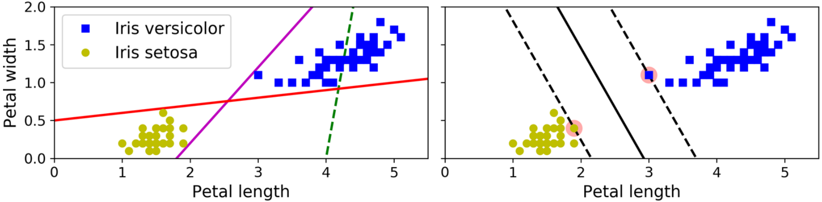

Image source: Géron, A. (2023), *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*, (3rd ed.)

SVM classifier finds the widest possible **gutter**. Instances located on the edge of the gutter are called **support vectors**. Distance from the support vector to the line is called the **margin**.

Linear classifiers and hyperplanes

Hyperplane: $w_0 + \bar{x}^\top \bar{w} = 0$.
The hyperplane divides the input space into two halves:
$\bar{x}_i : w_0 + \bar{x}^\top \bar{w} < 0$ and $\bar{x}_i : w_0 + \bar{x}^\top \bar{w} > 0$
The hyperplane is mapped to the *probability* that an instance belongs to a class.

Scaling

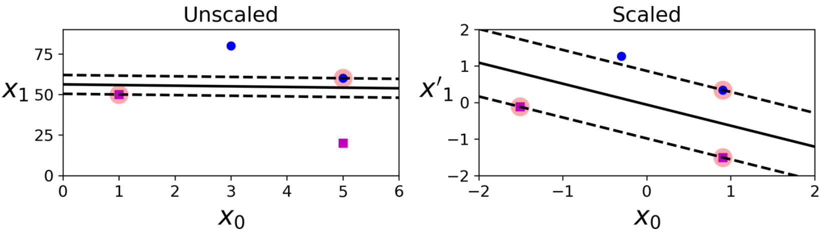

Image source: Géron, A. (2023), *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*, (3rd ed.)

SVM is vulnerable to **scaling**.

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Issue with outliers

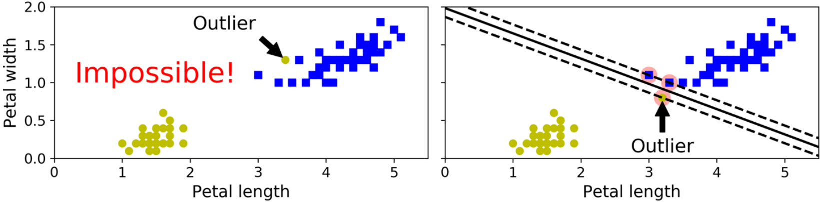

Image source: Géron, A. (2023), *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*, (3rd ed.)

Data is usually not linearly separable. SVM is highly **sensitive to outliers**. Instances in the gutter or on the wrong side are called **margin violations**.

Slack variable

Soft-margin SVM allow for slack variable ξ.
Values of ξ per instance:

- $\xi_i = 0$: on the correct side,
- $\xi_i > 0$: in the gutter (wrong side of margin),
- $\xi_i > 1$: on the wrong side of hyperplane.

Margin violations result in $\xi > 0$ (per instance).

Hyper-parameter C

C represents the amount that the margin can be violated.
The higher the value of C, the lower the slack, the less tolerant for violations to the margin.

- Small C: underfitting (tolerance for misclassification)
- Large C: overfitting (classify all correctly)

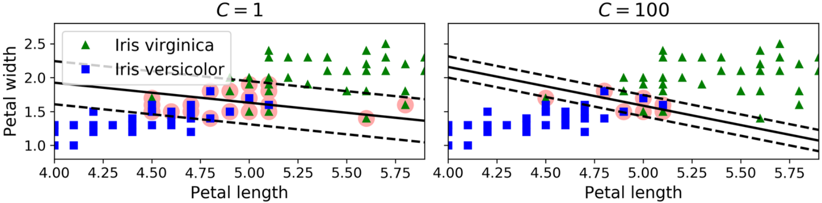

Image source: Géron, A. (2023), *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*, (3rd ed.)

Optimization problem

Soft margin SV classifier is the hyperplane solution to the optimization problem:

Minimize

$$\frac{1}{2}\|\bar{w}\|^2 + C\sum_{i=1}^{N}\xi_i$$

subject to

$$\xi_i \geq 0, \quad y_i\left(w_0 + \bar{x}_i^\top \bar{w}\right) \geq 1 - \xi_i, \quad i = 1, \ldots, N$$

Optimization problem

$$\frac{1}{2}\|\bar{w}\|^2 + C\sum_{i=1}^{N}\xi_i$$

- $\frac{1}{2}\|\bar{w}\|^2$: margin size (smaller $\bar{w}$ = larger margin),
- C: allowed margin violation. Generalized model (larger margin) vs. minimized misclassification errors (smaller margin),
- $y_i\left(w_0 + \bar{x}_i^\top \bar{w}\right) \geq 1 - \xi_i$: correctly classified points are at least margin 1 away from the hyperplane.

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Multiclass classification

= Non-binary classification

- One-vs-rest (OVR)
- One-vs-one (OVO)

One-vs-rest

"Is it Alice or not".

Training k SVMs. Prediction runs all SVMs and return the one with the best score.

One-vs-one

"Is it Alice or Bob". (disregard instances that are neither).

Training $\frac{k(k-1)}{2}$ SVMs (all combinations). Prediction runs all SVMs and return the one with the most common answer.

Exercise

Classification of 10 fruits. Should we use OvO or OvR? The training data includes 50 images of each fruit.

Number of models:

Size of training set for each:

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Non-linearly separable data

New dimension $x_2 = (x_1)^2$

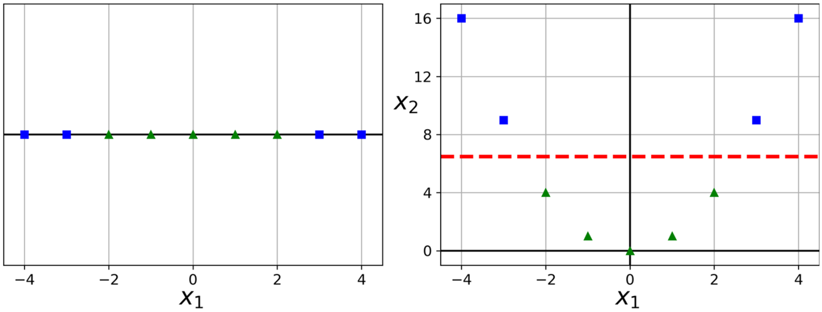

Image source: Géron, A. (2023), *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*, (3rd ed.)

Non-linearly separable shapes

Doughnut

Crescent Moons

Intertwined Spirals

Kernel function

Motivation: the more dimensions, the slower the algorithm.

Kernel function computes the similarity between two generic vectors $(\bar{x}_i, \bar{x}_j)$ in a transformed space $(\phi(\bar{x}_i))$ without computing the transformation.

$$K(\bar{x}_i, \bar{x}_j) = \phi(\bar{x}_i)^\top \phi(\bar{x}_j)$$

Linear kernel

Dot product of the two vectors:

$$K(\bar{x}_i, \bar{x}_j) = \bar{x}_i^\top \bar{x}_j$$

Equivalent to the standard SVM decision function:

$$f(\bar{x}) = w_0 + \bar{x}^\top \bar{w}$$

Polynomial kernel

Simulates adding many polynomial features without adding them. The degree of the polynomial d. ($d = 1$ is equivalent to linear kernel)

$$K(\bar{x}_i, \bar{x}_j) = (\gamma \bar{x}_i^\top \bar{x}_j + r)^d$$

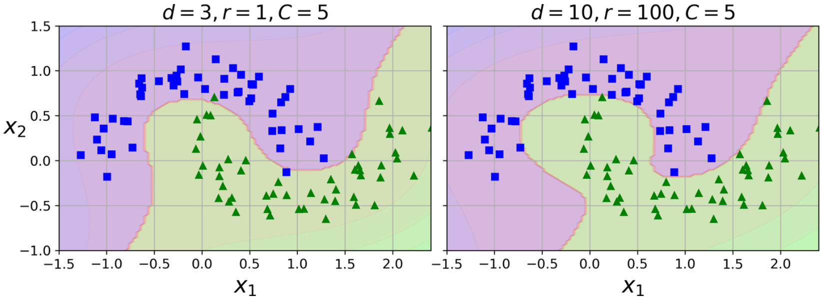

Image source: Géron, A. (2023), *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*, (3rd ed.)

RBF kernel = Radial Basis Function (Gaussian)

Maps the problem space into an infinite-dimensional space.
It is the default kernel in `sklearn`.

$$K(\bar{x}_i, \bar{x}_j) = \exp(-\gamma \|\bar{x}_i - \bar{x}_j\|^2)$$

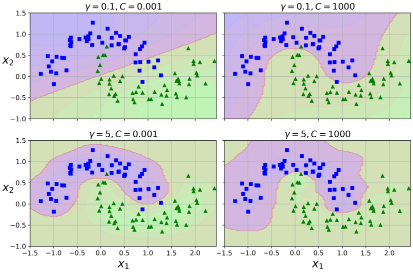

Image source: Géron, A. (2023), *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*, (3rd ed.)

Hyperparameters

- C: effects margin (same as for linear classifier)
- γ (gamma): curves decision boundaries
- d: complexity of the model (degree of polynomial). Mind differences between even and odd degrees.
- r (`coef0`): increases complexity when $r \neq 0$. Higher values give more weight to lower-degree terms of the polynomial.

Too high C and γ together results in overfitting (separate regions).

Comparison of kernel functions

Linear kernel:

- used when data is already linearly separable,
- fast performance,
- works well in high-dimensional spaces.

Polynomial kernel:

- maps data into higher-degree polynomials,
- allows for curved boundaries,
- performs faster than linear but slower than RBF.

RBF kernel:

- maps data into infinite-dimensional feature space,
- works well with complicated non-linear decision boundaries,
- slow performance.

Summary of SVM

Advantages:

- good performance on high-dimensional data,
- high flexibility,
- powerful technique.

Disadvantages:

- slow performance (if RBF kernel),
- vulnerability to scaling,
- might require pre-processing,
- requires hyperparameter tuning,
- might result in overfitting.

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Code

Github link: https://github.com/304125/MAL1_S26/tree/master/08_Support_Vector_Machines